



Digital rock physics benchmarks—part II: Computing effective properties

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ABSTRACT

This is the second and final part of our digital rock physics (DRP) benchmarking study. We use segmented 3-D images (one for Fontainebleau, three for Berea, three for a carbonate, and one for a sphere pack) to directly compute the absolute permeability, the electrical resistivity, and elastic moduli. The numerical methods tested include a finite-element solver (elastic moduli and electrical conductivity), two finite-difference solvers (elastic moduli and electrical conductivity), a Fourier-based Lippmann-Schwinger solver (elastic moduli), a lattice-Boltzmann solver (hydraulic permeability), and the explicit-jump method (hydraulic permeability and electrical conductivity). The set-ups for these numerical experiments, including the boundary conditions and the total model size, varied as well. The results thus produced vary from each other. For example, the highest computed permeability value may differ from the lowest one by a factor of 1.5. Nevertheless, all these results fall within the ranges consistent with the relevant laboratory data. Our analysis provides the DRP community with a range of possible outcomes which can be expected depending on the solver and its setup.

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1. Introduction

Over the past decade, the use of digital rock physics (DRP) for understanding pore-scale processes and estimating effective (macroscopic) rock properties has been increasing. DRP has combined technological advances from various allied engineering and scientific fields and effectively used it for earth-science related problems.

The DRP workflow primarily involves three steps (Dvorkin et al., 2011): (a) digitally imaging (e.g., high resolution 3D CT-scan) of small pieces of rocks to resolve their pore-scale features; (b) processing the raw images (gray-scale or intensity image) to separate pore from the mineral matrix phases and obtain a segmented image; and (c) simulating physical processes within this microstructural image to determine its effective properties, such as permeability (absolute and relative), electrical resistivity, and the elastic moduli (including the anisotropy constants if required). These 3D data-cubes are then used for simulating

physical processes. DRP methodology calls for high-resolution 3D imaging machines supplemented by image-reconstruction, processing and segmentation methods, as well as accurate process simulation computational engines.

As DRP advances for more wide-spread use in industry and academia, the need for establishing benchmark datasets and comparing existing numerical techniques on the same dataset becomes paramount. In the accompanying first paper of our series, we proposed four benchmark datasets: Fontainebleau sandstone, Berea sandstone, Grosmont carbonate, and a spherical bead pack, along with necessary acquisition and processing details. In this final part, we concentrate on computing the effective properties using several process-simulating computational methods on the same digital rock datasets. The physical properties we have concentrated on are: single-phase hydraulic permeability, electrical resistivity (or its inverse, conductivity), and the isotropic elastic moduli.

In subsequent sections, we present concise descriptions of process simulation computational engines for elastic, transport, and electrical properties. We give an overview of the fundamental equations and boundary conditions being used. Then, we discuss key similarities and differences between these methods, and give an overview of the fundamental equations and boundary conditions.

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2. Numerical methods for computing effective properties

The third critical step in the digital rock physics workflow is to simulate physical processes in the segmented pore-space and mineral-matrix and thereby obtain volume-averaged physical property estimates. Various algorithms have been proposed over the last two decades for simulating these micro-scale physical processes and estimating macro-property values. In this benchmarking project, we use a selection of established as well as new methods. Two popular methods for computing static elastic and electrical properties from digital rock images are the finite-element and finite-difference methods (Garboczi, 1998; Arns et al., 2002; Xuefeng et al., 2009; Zhan et al., 2010). An established method to compute pore-scale fluid flow simulation is the lattice-Boltzmann method (Ferreol and Rothman, 1995; Keehm, 2003; Fredrich et al., 2006). In addition to the aforementioned methods, we present here the dynamic pulse propagation approach for computing elastic properties (Saenger, 2008), as well as two new methods which have not been applied in the field of digital rock physics, yet. Those are the explicit-jump immersed interface method (Wiegmann, 2007) for computing transport properties and an elastic solver based on the Lippmann-Schwinger equation (Moulinec and Suquet, 1998). Both methods make use of the fast Fourier transform.

A common feature of all the methods discussed in this paper is a discretization of the underlying partial differential equations on a regular Cartesian grid, which is directly derived from the segmented digital rock voxel image. The generation of a volumetric irregular mesh is therefore not necessary.

2.1. Elastic property estimators

We use three different approaches for computing effective elastic moduli: the dynamic pulse propagation method, the variational approach using finite elements, and an integral formulation based on the Lippmann-Schwinger equation. Depending on the numerical set-up, these approaches can also provide the full stiffness tensor. One key distinction between these different solvers is dynamic versus static approaches. It should be noted that these approaches give estimates of elastic moduli and do not account for non-linear or pressure-dependent rock behavior. Hence, the static moduli thus computed should not be confused with inelastic moduli from large-strain static experiments in geomechanics and engineering.

Assuming a linear elastic regime, the elastic displacement field u is described by the momentum equations

$$\rho \ddot{u} = \nabla \cdot \tau, \quad (1)$$

where ρ is the material density and the stress field $\tau(u)$ can be computed from the elastic strain field ϵ and the stiffness tensor C using Hooke's law $\tau = C : \epsilon$. We differentiate between dynamic and static methods depending on whether the field equations include the inertial term $\rho \ddot{u}$ or not.

Computing the static stress field within a heterogeneous medium is a boundary value problem that requires the specification of boundary conditions. Examples of boundary conditions are: (a) Dirichlet boundary conditions; (b) Neumann boundary conditions; and (c) periodic boundary conditions. Dirichlet and Neumann conditions refer to the specification of displacements and stresses, respectively, at all exterior boundaries of the domain. The periodic condition implies that all physical fields, stresses and displacements, are the same at any corresponding pair of points that is located at opposing sides of a cuboidal modeling domain. Periodic boundary conditions appear naturally in numerical methods that make use of the Fourier transform.

The application of Dirichlet and Neumann boundary conditions give values closer to the upper bound and the lower bound

Table 1

Mineral properties used for the computation of effective elastic properties.

	Density (g/cm ³)	Bulk modulus (GPa)	Shear modulus (GPa)
Pore space	0.00	0.0	0.0
Quartz	2.65	37.0	44.0
Calcite	2.71	68.3	28.4
Dolomite	2.87	94.9	45.0

of the composite-media elastic properties, respectively (Saenger et al., 2006), while periodic boundary conditions can provide estimates which fall between the upper and lower limits. In the case of inclusions with dilute concentrations, the periodic boundary condition yields estimates that are closest to theoretical estimates (Garboczi and Berryman, 2001).

An alternative to static elastic simulation is the pulse-propagation method (dynamic numerical simulation). This method is based on the dynamic momentum equations, such that in addition to the boundary conditions, initial conditions need to be specified.

A list of the elastic material properties used in our numerical simulations is given in Table 1.

2.1.1. Dynamic approach for dry rock moduli

The dynamic pulse propagation method uses a rotated-staggered-grid finite difference approach for modeling elastic wave propagation (Saenger et al., 2000; Saenger, 2008). The basic idea of this approach is to study speeds of elastic waves through heterogeneous materials in the long wavelength limit (pore size $\ll$ wavelength). In addition to the determination of dry rock moduli, this method is also designed to study frequency-dependent effects caused by viscous pore fluids (Saenger et al., 2011).

For the purpose of estimating effective elastic properties, the digitized rock sample is embedded in a homogeneous elastic region. We assign elastic properties of the grain material to this region. We perform our experiments with periodic boundary conditions in the directions parallel to the wave propagation. A body-force plane-source is applied at the top of the model, where the source signal is a broadband Gaussian pulse. The plane P- or S-waves generated in this way propagate through the numerical model. With two planes of receivers at the top and at the bottom of the model, it is possible to measure the time-delay of the peak amplitude of the mean plane wave caused by the inhomogeneous structure of the digitized rock sample. With the time-delay (compared to a reference model) we estimate the effective velocity of both the compressional and the shear waves, and hence, the effective rock moduli (e.g., Saenger et al., 2011). In the following, we refer to the pulse propagation method as dynamic finite-difference method (FDM-dynamic).

2.1.2. Finite-element method for static elasticity

We use an implementation of the finite-element method, which is a well-established method of numerically computing elastic properties for two-dimensional and three-dimensional images (Roberts and Garboczi, 2002a, 2002b; Garboczi and Berryman, 2001; Meille and Garboczi, 2001; Keehm, 2003; Garboczi et al., 1995). The numerical computations consider each voxel as a tri-linear element, so that the entire digital lattice can be treated as a finite element mesh (Garboczi and Berryman, 2001). Further, periodic boundary conditions are assumed for the microstructures. The elastic displacements are linearly extrapolated across all voxels, and a variational formulation of the linear elastic equations is imposed. This implementation finds the solution by minimizing the elastic energy using a fast conjugate-gradient method (Arns et al., 2002). The effective moduli are usually

defined by a stress average. Garboczi and Berryman (2001) and Roberts and Garboczi (2002a) discuss the main sources of error in these computations: (a) finite-size effect; (b) digital resolution; and (c) statistical variation. In this work, we use two different implementations (in Fortran and C++) of the basic finite element code which will be denoted as FEM-F and FEM-C.

2.1.3. Integral equation and Fourier transform method

We use an implementation of a numerical method introduced by Moulinec and Suquet (1998) based on the Lippmann-Schwinger equation (Zeller and Dederichs, 1973) allowing the computation of local and global elastic response of composites.

For the homogenization of linear elastic properties, a uniform macroscopic strain field $\mathbf{E}$ is prescribed on the cuboidal domain Ω of the medium. The minimization of the strain energy is equivalent to a boundary value problem (BVP) consisting of the elastic equilibrium equation, Hooke's law, and periodic boundary conditions. This BVP is transformed into an equivalent domain integral equation using the concept of background stress polarization and by expressing the local strain field with the help of the elastic Green operator Γ_0 . Thus, one arrives at the Lippmann-Schwinger equation

$$\boldsymbol{\varepsilon} + \Gamma_0 * ([\mathbf{C} - \mathbf{C}_0] : \boldsymbol{\varepsilon}) = \mathbf{E} \quad (2)$$

where $\mathbf{C}_0$ denotes a well-chosen homogenous reference material, and the star symbol is the convolution operator. Eq. 2 with corresponding boundary conditions is equivalent to the original elastic BVP. The equation is solved iteratively by the Neumann series expansion. The application of Γ_0 is done in the Fourier space and the Fourier transformations by the fast Fourier transform (FFT).

The discretization method is well adapted for image-based simulation of microstructure properties. The Lippmann-Schwinger (L-S) method has two main advantages: (1) it directly computes the local strain and stress as primary unknowns, which are needed for the averaging (strain, stress, or strain-energy equivalence principles) in the homogenization method; (2) the method is fast and efficient with respect to memory consumption due to the use of the FFT.

2.2. Single-phase fluid flow

For last few decades, several studies have estimated permeability from digital images of rocks. Permeability depends on pore geometry and pore size. The work in this field can be generally divided into two groups. The first group involves estimating different parameters, including surface area and grain size distribution, which are used in semi-empirical models like the Kozeny-Carman relation to compute permeability. These relations are empirical due to the scale factors that are used to fit the data. In this paper, we used the second category of approaches which numerically solve the Navier-Stokes equation for low-Reynolds-number fluid flow in the pore. For a review of finite-difference, finite-elements, lattice-gas automata and Lattice-Boltzmann methods for fluid flow in porous media, we refer to Keehm (2003).

In this study, we compare two methods for computing the hydraulic permeability. These are the Lattice-Boltzmann method (LBM) and an EJ Stokes solver (EJ). Both methods solve the Stokes equations

$$-\mu \nabla^2 \mathbf{u} + \nabla p = \mathbf{f} \quad (3)$$

$$\nabla \cdot \mathbf{u} = 0 \quad (4)$$

within the rock microstructure. Here, p is the pore fluid pressure, $\mathbf{u}$ the velocity vector and μ denotes the dynamic fluid viscosity. With the solution of the Stokes equations the permeability tensor

$\mathbf{K}$ is calculated using Darcy's law

$$-\mathbf{u} = \frac{\mathbf{K}}{\mu} : \nabla p. \quad (5)$$

To calculate the full tensor, three flow calculations, in the x-, y- and z-directions, respectively, are needed.

2.2.1. Lattice-Boltzmann method

The Lattice-Boltzmann method is a powerful technique for computational modeling of a variety of complex fluid flow problems, including single and multi-phase flow in complex geometries (Martys et al., 1999). The main advantage of using LBM is its usability for very complex pore-geometries of rocks (Cancelliere et al., 1990, Ladd, 1994a, Martys and Chen, 1996, Keehm, 2003). This method does not idealize the pore-space and hence provides a rigorous estimate of flow properties. Further, LBM can give an accurate solution for flow properties in agreement with finite-element simulation (Kandhai et al., 1998, 1999). This method is based on cellular automata theory and describes the fluid volume in a complex pore-geometry in terms of the interactions of a massive number of particles following simple local rules (Keehm, 2003, Doolen, 1990, Chopard and Droz, 1998). The rules in this method recover the Navier-Stokes equation at the macroscopic level (Rothman and Zaleski, 1997, Ladd, 1994b).

Here, we use an implementation of LBM (Keehm, 2003) to simulate single-phase flow with a "fixed flux" scheme and 15 buffer layers at the inlet and the outlet. The absolute permeability is estimated by calculating the inlet and the outlet pressures and applying them to Darcy's law.

2.2.2. Explicit jump Stokes method

One method used to calculate the permeability from rock microstructure images is the Explicit Jump (EJ) method (Wiegmann and Bube, 2000). The Stokes equations are solved on an equidistant Cartesian grid on a box-shaped domain for a given pressure difference in the x-, y- and/or z-direction. The boundary conditions are periodic, meaning that pressure is periodic with a discontinuity on the boundary in the direction under examination. All three components of the velocity are periodic in the usual sense, i.e. the fluxes at corresponding points at the inlet and the p outlet are equal.

The staggered grid approach from Harlow and Welch (Harlow and Welch, 1965), with no-slip boundary conditions on the voxel faces, is used to discretize the equations. The discretized equations are solved with a fast Fourier method similar to the FFF-Stokes method (fast fictitious forces, Wiegmann, 2007). The EJ-method improves the FFF-method, in using Harlow and Welch's accurate boundary conditions for the tangential variables rather than a fictitious force.

2.3. Direct current electrical conductivity

The effective electrical conductivity of a rock sample depends on the conductivities of the grain minerals and the pore-filling fluids. In most sedimentary rocks, the solid grain materials have conductivities that are orders of magnitude lower than the conductivity of the pore fluid and in this case the effective properties are fully determined by the fluid properties, the sample's porosity and the connectivity of the pore space.

In the static direct current limit, the electrical current density field is described by a diffusion equation for the electrical potential u

$$0 = \nabla \cdot (\sigma \nabla u). \quad (6)$$

In this equation, $\sigma(x)$ denotes the local electrical conductivity. We assume that the solid grains are insulators, and therefore assign conductivity values of 0 in the solid matrix and a fluid conductivity σ_{fluid} in the pore fluid. The effective conductivity is then obtained using Ohm's law relating the total electrical current

$I = \sigma_{eff} U$, where U denotes the voltage between the inlet and the outlet boundaries. Following common practice, we present the conductivity in terms of the formation factor $F = \sigma_{fluid} / \sigma_{eff}$.

We compare three numerical methods, the finite-difference, finite-element, and explicit jump diffusion methods for computing the effective electrical conductivity of porous rock samples.

2.3.1. Finite-difference method

We use an optimized staggered-grid finite difference solver with 2nd order accuracy in space (Garboczi et al., 1998; Schwartz et al., 1989; Zhan et al., 2010). In order to handle high contrast conductivity values for neighboring grids, we adopt a gradual relaxation method (Press et al., 2007). For calculations on the sphere pack, carbonate and sandstone images, periodic boundary conditions are applied using imaginary layers in all directions.

2.3.2. Finite-element method

The numerical solution of the electrical diffusion equation is performed using an implementation of the finite-element method by the National Institute of Standards and Technology (Schwartz et al., 1989). The discretization uses a regular voxel grid where each voxel is mapped to a linear cubic element. The solver applies periodic boundary conditions. For the digital rock samples of sandstone and carbonate, the model is mirrored in the direction of the applied voltage, whereas no mirroring is performed in the perpendicular directions. No mirroring is necessary in the case of

the sphere pack, as the corresponding model geometry is periodic. As an alternative to the sample mirroring, a conducting layer can be incorporated at the downstream boundary of the sample geometry, ensuring a connected flow path in the presence of the periodicity condition.

2.3.3. The EJ-method for effective electrical conductivity and diffusivity

The third method to calculate the effective electrical conductivity is the Explicit-jump diffusion method. It solves the diffusion eq. (6) on an equidistant Cartesian grid on a box-shaped domain for a given difference in the potential in the x-, y- and/or z-direction. The boundary conditions are periodic, meaning that the potential is periodic with a discontinuity on the boundary in the direction under consideration, in analogy to the EJ-Stokes solver. The EJ-method to solve conductivity problems is described in Wiegmann and Zemitis (2006).

3. Results

We compute the bulk and shear moduli, hydraulic permeability, and electrical resistivity by the aforementioned methods for four rock samples and analyze the variability of the numerical predictions. As described in the preceding section, the numerical methods for elastic properties are the Fourier-transformed based Lippmann-Schwinger

Table 2

Simulation results for elastic and transport properties. The methods are the Lippmann-Schwinger method (L-S), the finite-element method (FEM-C, FEM-F), the dynamic finite-difference pulse propagation method (FDM-dyn.), lattice-Boltzmann method (LBM), the Explicit Jump method (EJ), and the finite-differences method (FDM).

Property	Method	Fontb.	Berea	Berea	Carb.	Carb.	Spheres
Image size	x	288	400	724	400	1024	788
	y	288	400	724	400	1024	791
	z	300	400	1024	400	1024	793
Porosity		0.147	0.184	0.184	0.247	0.247	0.343
Bulk modulus (GPa)	L-S	22.6	19.2	19.1	24.8	25.0	5.0
	FEM-C	24.1	19.9	19.1*	22.7	23.7*	3.8
	FEM-F	24.3	20.1	20.3*	22.5	22.0*	3.5
	FDM-dyn.	23.6	18.6	19.8**	17.3	21.5**	
	Midrange M	23.4	19.4	19.7	21.0	23.2	4.2
	Range R	1.7	1.5	1.2	7.5	3.4	1.5
	½ R/ M (%)	3.6	4.0	3.1	17.7	7.4	17.9
	L-S	29.1	23.6	23.5	14.0	13.6	5.1
	FEM-C	26.6	20.5	19.5*	12.2	13.1*	4.0
	FEM-F	26.6	20.7	21.1*	12.6	12.4*	3.7
Shear modulus (GPa)	FDM-dyn.	23.5	20.6	19.9**	12.3	11.5**	
	Midrange M	26.3	22.1	21.5	13.1	12.6	4.4
	Range R	5.6	3.1	4.0	1.8	2.2	1.4
	½ R/ M (%)	10.7	7.1	9.2	6.7	8.7	15.6
Permeability (mD)	EJ stokes	1914	124	139	512	1706	270220
	LBM	1503	113	109*	353	773*	221587
	Midrange M	1708	118	124	433	1239	245903
	Range R	411	10	30	158	933	48633
	½ R/ M (%)	12.0	4.3	12.2	18.3	37.6	9.9
	EJ diffusion	41.6	22.5	24.8	19.5	20.9	4.6
	FEM-C	29.4	22.0	20.9*	16.9	22.6*	4.8
	FEM-F	29.6	20.2	21.6*	16.6	16.9*	4.3
	FDM		29.7	30.9	24.8	22.3	4.7
	Midrange M	35.5	25.0	25.9	20.7	19.8	4.5
Formation factor	Range R	12.2	9.5	10.0	8.2	5.7	0.4
	½ R/ M (%)	17.1	19.0	19.3	19.9	14.3	4.8

*Result obtained from arithmetic averaging of subsamples. **Result obtained from coarsened sample with a resampling factor of 2.

method (L-S), two implementations of the finite-element method, referred to as FEM-F and FEM-C, as well as the dynamic finite-difference pulse propagation method (FDM-dynamic). Hydraulic permeability is estimated using the Explicit Jump immersed interface method (EJ Stokes) and the Lattice Boltzmann method (LBM). The solvers used for the electrical resistivity computations are the Explicit Jump Diffusion method (EJ Diffusion), two implementations of the finite-element method for electrical conductivity (FEM-C and FEM-F), as well as the finite-difference method (FDM).

All ten solvers are used to predict the effective physical properties for each of the four benchmark samples. For the Fontainebleau sandstone and the sphere pack, we use a single segmented image per sample. Simulations of the Berea sandstone and the Grosmont carbonate are first carried out using a specific segmented image that we refer to as “VSG reference segmentation”. Then, we apply the LBM method and the FEM-C method to a second segmented image (“Kongju segmentation”). The FEM-F solver is used to compute elastic and electrical properties using a third segmented image (“Stanford segmentation”).

The dimensions of the Fontainebleau sandstone, the Berea sandstone, the carbonate, and the sphere pack images are $300 \times 300 \times 288$, $1024 \times 1024 \times 724$, $1024 \times 1024 \times 1024$, and $788 \times 789 \times 791$ voxels, respectively. Due to limited computational resources, most simulation results are performed on smaller subsamples that we extracted from the larger volume. Simulations corresponding to the LBM and the FEM-C solvers are run using subvolumes of the size $400 \times 400 \times 400$. For the Berea sample, two subsamples were used, for the carbonate sample, 8 subsamples were used and for the sphere pack, 1 subsample was used. The Berea and

carbonate results using the FEM-F solver are based on subsamples with the dimensions of $724 \times 724 \times 512$ voxels, and for both rock types, two subsamples were considered. Simulation results corresponding to the EJ method, the FDM method, the dynamic pulse propagation method, and the L-S method for elastic properties were performed using the full model sizes. The “full-scale” values thus obtained are compared to simulation results obtained using subsamples by applying an arithmetic average to the computed values of the subsamples. The variability in numerical results obtained for subsamples of the same rock image reflect the rock heterogeneity.

In addition to the full-scale Berea and carbonate samples, we provide simulation results on a smaller $400 \times 400 \times 400$ voxel subsample that is chosen such that the porosity of the subsample is identical to the porosity of the full sample. In the electronic supplements to this paper, we report the locations of all subsamples within the respective larger volumes, as well as the corresponding simulation results.

Transport properties are computed in the three Cartesian directions, and we compute “isotropic” equivalent values by arithmetically averaging the directional permeabilities and conductivities, respectively. Similarly, we report equivalent isotropic elastic moduli.

An overview of the simulation results is given in Table 2. All computational results, i.e. including the values obtained on subvolumes, are shown in Figs. 1–8. We plot elastic and transport properties in two plot sets. The first set, Figs. 1, 3, 5, and 7, shows computations made on images segmented by the reference segmentation algorithm. The second set, Figs. 2, 4, 6, and 8, shows the same data, complemented by results that are obtained using different segmented images of the Berea and the carbonate

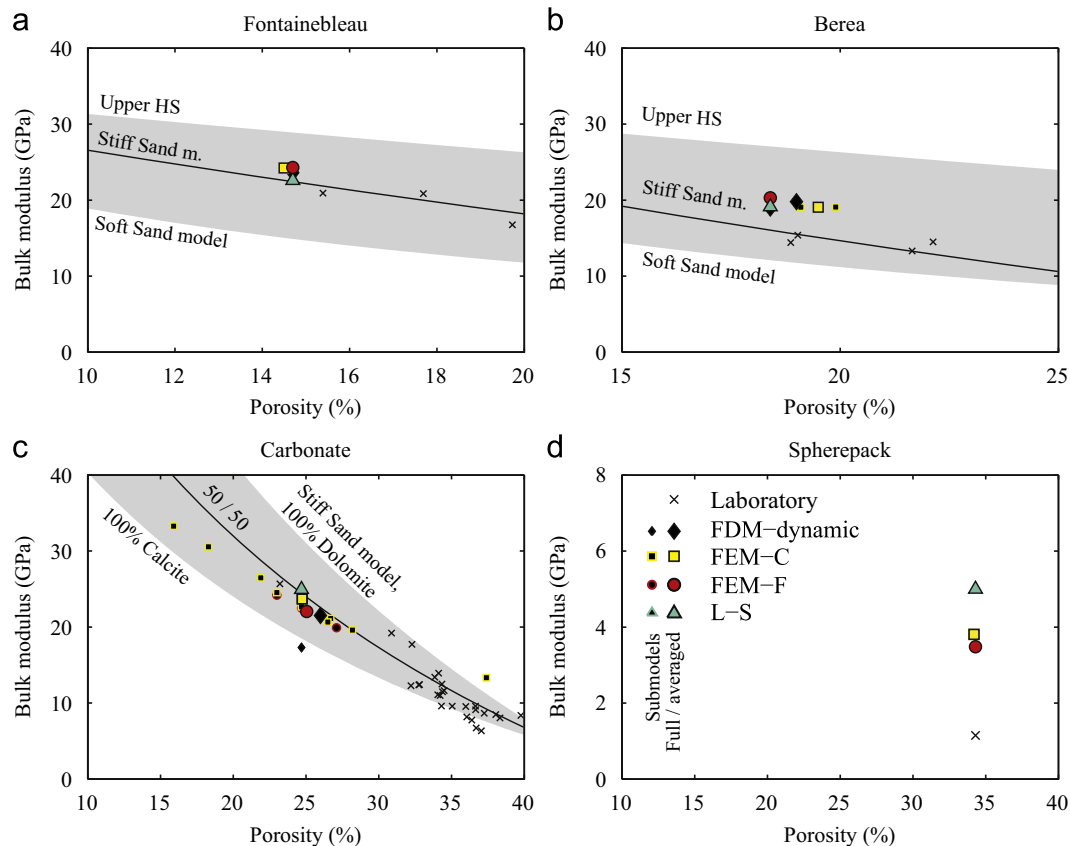


Fig. 1. Bulk moduli of all samples computed for the reference segmentations: (a) Fontainebleau sandstone, (b) Berea sandstone, (c) Carbonate sample, (d) Binarized sphere pack. Laboratory data points are derived from Han's (1986) Fontainebleau and Berea dry measurements at 40 MPa, Carbonate derived from velocity data of Vanorio et al. (2008). The gray-shaded areas correspond to the upper and modified lower Hashin-Shtrikman bounds (soft sand model) for the sandstones, the solid line represents the so-called stiff sand model (Dvorkin and Nur, 1996). We model the carbonate data using the stiff sand model using different calcite, dolomite and a mixture mineral composition. The “laboratory” data point for the sphere pack is a reference solution from granular dynamics simulation.

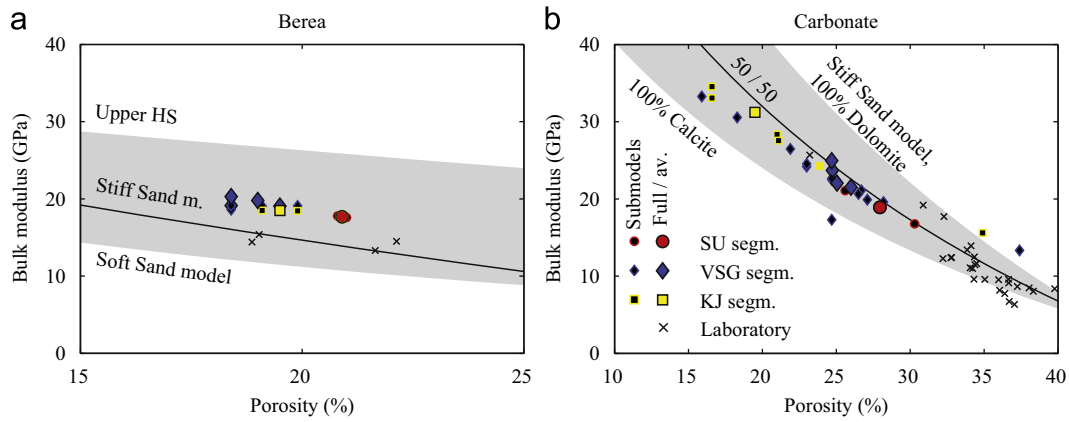


Fig. 2. Bulk moduli of Berea and carbonate samples computed for all segmentations.

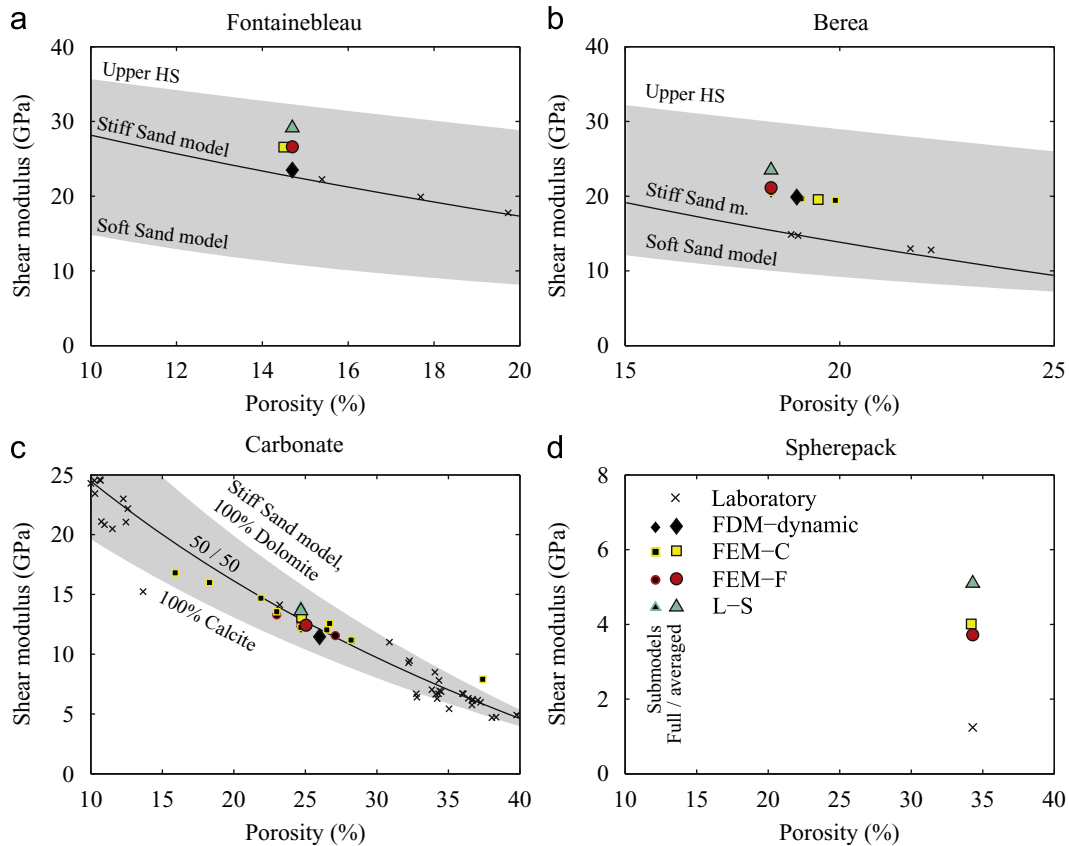


Fig. 3. Same as Fig. 1, but for the shear modulus.

samples. We provide relevant laboratory data for all computed properties. Note, however, that the laboratory data was not measured using the same plugs from which the CT scans were made, nor were they performed on sample sizes comparable to those used during the CT imaging process. The sandstone laboratory measurements were obtained from various Berea and Fontainebleau samples (Yale, 1984; Han et al., 1986; Gomez, 2009), the carbonate dataset was obtained from a rock formation in Apulia, Italy, which has similar microstructural characteristics as the Grosmont carbonate under examination (Vanorio et al., 2008).

3.1. Bulk and shear moduli

Simulated elastic properties are given in Figs. 1–4. see Table 2. The elastic moduli predicted using different solvers reveal

variability. The best agreement is observed for the bulk modulus of the Berea sandstone, where the maximum deviation from the midrange value of all predicted bulk moduli is 0.6 GPa.

The elastic moduli computed for the sphere pack vary between 3.5 GPa and 5.0 GPa for the bulk modulus, between 3.7 GPa and 5.1 GPa for the shear modulus. This corresponds to a relative deviation of 18% and 16% from the respective midrange value. It is the largest relative disagreement of all simulated samples.

The eight subsamples of the carbonate sample computed using the FEM-C solver show a large range of porosities, reflecting the heterogeneous nature of the sample. The corresponding elastic results form a linear trend which is consistent with the other simulations both for the bulk and the shear modulus (see Figs. 1c and 2b).

The L-S method tends to predict higher elastic moduli (in particular the shear moduli) as compared to the other methods,

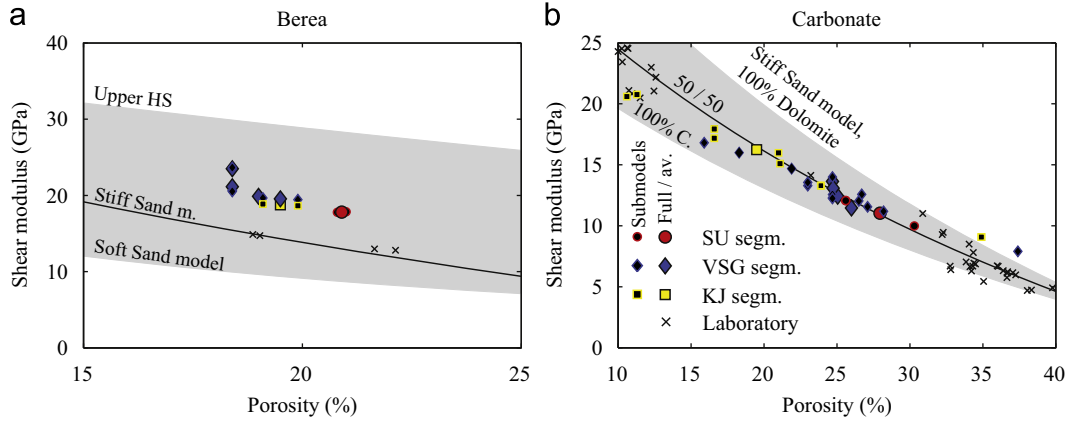


Fig. 4. Shear moduli of Berea and carbonate computed for all segmentations.

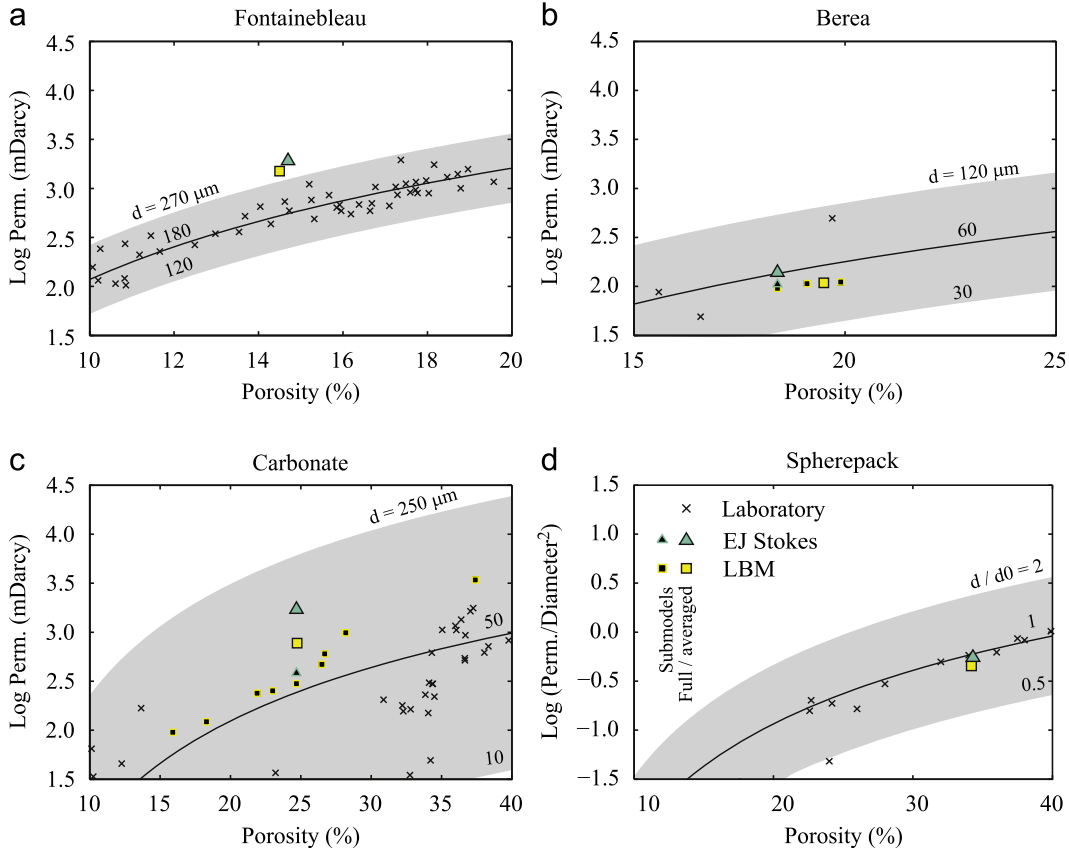


Fig. 5. Permeability of all samples computed for the reference segmentations: (a) Fontainebleau sandstone, (b) Berea sandstone, (c) Carbonate sample, (d) Binarized sphere pack. The gray shaded area and the line represent a Kozeny Carman trend for different grain sizes (critical porosity $\phi_c = 0.035$). Laboratory data points for Fontainebleau and carbonate are taken from ultrasound measurements by Bourbie and Zinszner (1985); Gomez (2009). Berea data was measured by Yale (1984). Sphere pack permeability data was measured on sintered glass beads (Guyon et al., 1986).

while the dynamic pulse propagation method tends to produce relatively lower values.

Figs. 2 and 4 are the same as Figs. 1 and 3, respectively, except that in the former we add complementary data from simulations that use different segmented images. There is a significant variability of estimated porosities depending on which segmentation algorithm is applied to the gray-scale image. This variability is reflected in strongly variable elastic moduli which are estimated based on those segmented images. However, the simulation results obtained using the additional segmented images are consistent with those simulations using the reference images of

both Berea sandstone and the carbonate, respectively: The simulated elastic moduli align along a linear trend, see Figs. 2 and 4.

As a reference, we provide elastic laboratory data derived from ultrasonic experiments (Han, 1986; Vanorio et al., 2008, Gomez, 2009) and a reference solution for the sphere pack which has been computed using granular dynamics simulation (Sain, 2010). We observe good agreement for the case of Fontainebleau sandstone (Figs. 1a and 3a) and a carbonate dataset (Figs. 1c and 3c). The simulated elastic moduli for the sandstones and the sphere pack are higher than the corresponding laboratory and reference solutions; see Figs. 1b, 3b, 1d and 3d.

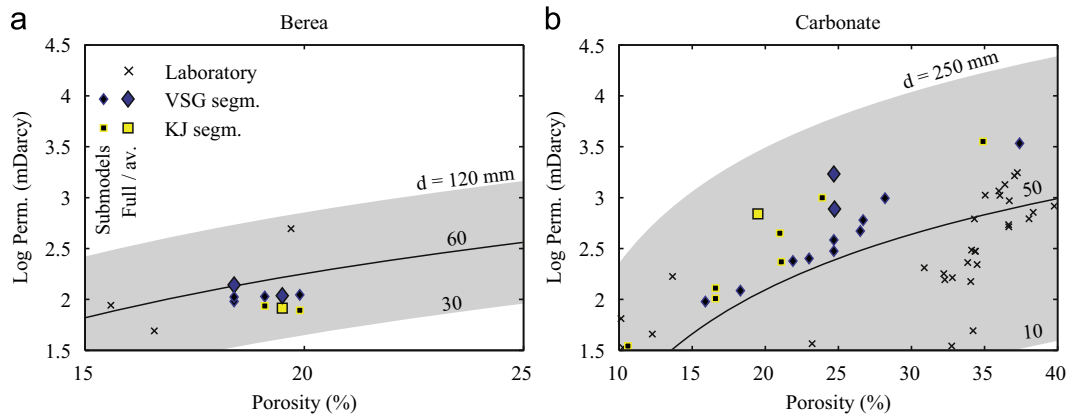


Fig. 6. Permeability of Berea and carbonate computed for all segmentations.

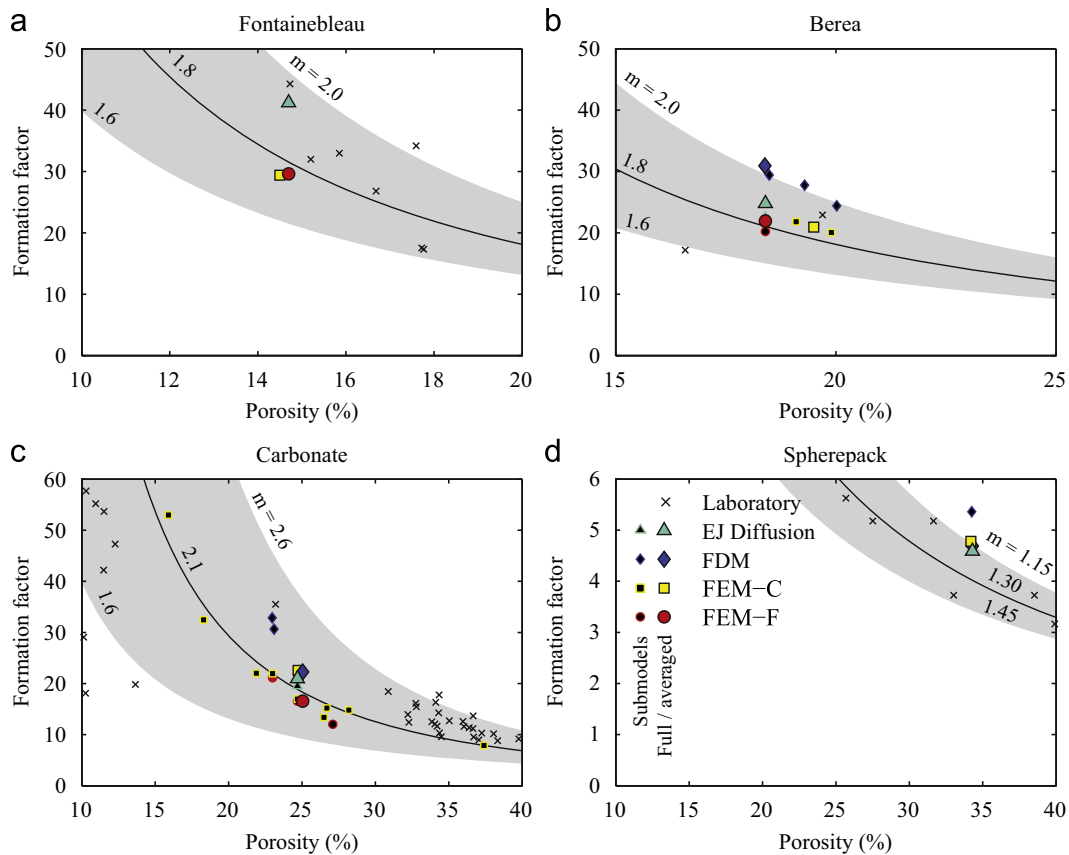


Fig. 7. Formation factor of all samples computed for the reference segmentations: (a) Fontainebleau sandstone, (b) Berea sandstone, (c) carbonate sample, (d) binarized sphere pack. The gray shaded area and the solid line represent Archie's law with different cementation factors. Laboratory data for Fontainebleau and carbonate are taken from Gomez (2009), from Yale (1984) for the Berea sandstone, Sen et al. (1981) for the sphere pack. One data point for Fontainebleau is from Fredrich et al., (1993).

All simulated data are well within the elastic bounds formed by upper and modified lower Hashin-Shtrikman bounds, the latter often referred to as the soft sand model. We assume pure quartz for both the Fontainebleau and the Berea sandstone, as well as a critical porosity of 0.4. As a reference, we also show the so-called stiff sand model (Dvorkin and Nur, 1996). For the details about these rock physics models, readers are referred to Mavko et al. (2009).

The elastic moduli of carbonate are modeled using the stiff sand model with a mineralogical mixture consisting of dolomite and calcite. Elastic properties for both minerals are provided in Table 1. We assume a critical porosity of 0.5 for modeling the carbonate rock.

3.2. Permeability

Estimated hydraulic permeability results are shown in Figs. 5 and 6. Results for natural rock samples (Fig. 5a–c) reveal that the LBM results are consistently providing the smallest permeability estimates, the solution using the Stokes EJ being slightly higher. There is very good agreement between both methods in the case of the Berea sandstone subsample. Depending on the model size, the two methods under examination predict permeabilities that differ by 10 or 30 mD, where the midrange value is 118 or 124 mD, respectively. The carbonate sample shows larger disagreement between the two methods.

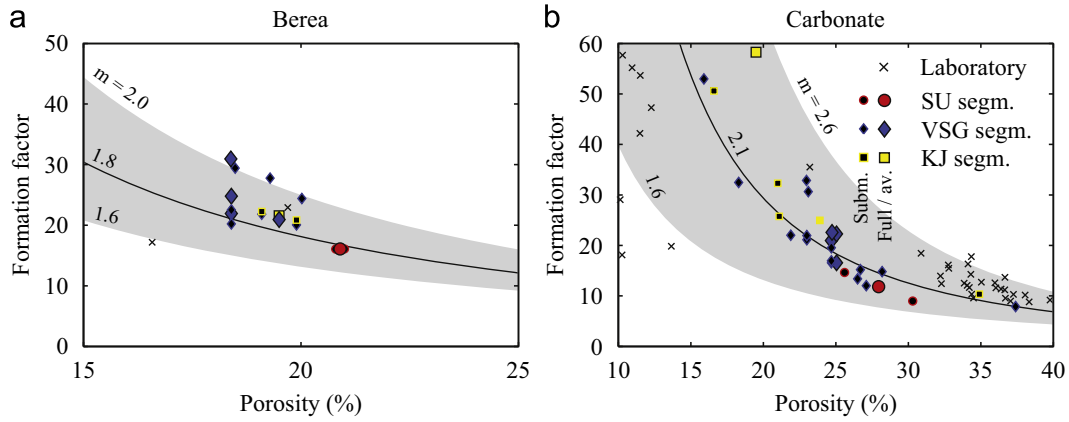


Fig. 8. Formation factor of Berea and carbonate computed for all segmentations.

In the case of the full-scale model, LBM predicts 773 mD, whereas the EJ Stokes method predicts 1706 mD.

Numerical estimates of permeability of the sphere pack are in very good agreement with laboratory measurements of sintered glass beads (Guyon et al. 1986; Gomez, 2009). Sphere pack results have been normalized with the square of the sphere diameter. The Berea sandstone permeability is consistent with measurements conducted by Yale (1984). Comparing the results for Fontainebleau with laboratory data reveals that all simulations predict values above those measured in the laboratory. Similarly, in the case of the carbonate, computed permeabilities are generally larger than laboratory measurements, regardless of the segmentation and solver algorithms.

3.3. Electrical resistivity

Electrical resistivity results are presented in terms of the formation factor and are given in Figs. 7 and 8. The gray-shaded areas correspond to modeled electrical resistivities using Archie's law

$$F = \phi^{-m} \quad (7)$$

where the range of available cementation factors m corresponds to the range of available laboratory data. We choose $m = 1.6; 1.8; 2.0$ for Berea/Fontainebleau sandstone, $m = 1.6; 2.1; 2.6$ for the carbonate rock and $m = 1.15; 1.30; 1.45$ for the sphere pack.

For the Fontainebleau sample (Fig. 7a), three simulation results are available from the two implementations of the finite-element method and for the EJ Diffusion method. We observe a range of the predicted formation factor between 29.4 and 41.6, which corresponds to a maximum deviation of 17% from the midrange value. For the Berea sandstone and carbonate samples, we find a maximum relative deviation of 20% of the midrange value.

High agreement between the EJ, the FDM, and the FEM methods is observed for the sphere pack. All simulation results are between 4.3 and 4.8, meaning that the largest relative deviation from the midrange value is less than 5%. Those values are also in good agreement with laboratory measurements reported for glass beads (Sen et al., 1981).

The FDM method tends to predict higher values if compared to the other methods. The FEM-F method shows the lowest values of electrical resistivity. Laboratory data of electrical resistivity is scarce but we observe in general good agreement with reported values (Gomez, 2009; Yale, 1984; Sen et al., 1981).

We note that in the case of the carbonate samples, the reported values corresponding to the FEM-C method show a very large range of porosities and electrical resistivities. The relationship between both properties is highly non-linear. In Table 2, we provide arithmetically averaged values of the formation factor.

Caution is necessary when comparing simulation results that were obtained from the averaging of multiple subsamples.

4. Discussion

We observe a variability of predicted effective properties in our numerical experiments. Within the digital rock physics workflow, we identify three factors which influence the numerical predictions:

1. The segmentation algorithm,
2. The solver algorithm and boundary conditions,
3. The subsample size and inherent rock heterogeneity.

With these three factors in mind, we analyze our computationally obtained properties (bulk modulus, shear modulus, single-phase permeability and formation factor) for the four benchmark digital datasets: Fontainebleau sandstone, Berea sandstone, carbonate, and a spherical bead pack. In the following, we discuss how much the aforementioned factors contribute to the variability in simulation results.

We should re-iterate the fact that the laboratory measurements used in our plots are not from the same sample as the ones used for numerical simulations. Also, the laboratory measurements are performed on core samples, while the DRP predictions are obtained from much smaller samples of the scale of several millimeters. It is known that even relatively homogeneous rocks have properties that vary on the scale of centimeters. Hence, it is not possible to directly compare laboratory results with our numerical predictions. The purpose of the laboratory measurements reported here is to demonstrate the qualitative agreement between simulated results and laboratory measurements, and show similarity in data-trends.

4.1. Segmentation

All considered rock properties are strongly correlated with the sample porosity, and, therefore, the choice of the segmentation algorithm impacts the computed effective properties. Some properties are computed using the same numerical solver and two different images segmented using the same gray-scale dataset. Those experiments reveal that the thus computed effective properties are consistent with one another in the sense that they form simple trends. These trends correspond to porosity-property relationships observed in laboratory data. If those trends are

taken into account, the relative importance of the segmentation process is smaller than if we consider absolute values.

4.2. Solver algorithm and boundary conditions

Numerical algorithms provide approximate solutions to the problem of rock physics property upscaling. The choice of numerical discretization and of the boundary conditions used for a digital experiment can affect the computationally obtained property estimates. In our benchmark results, we note several examples in which the simulation results of digital sample properties vary depending on the numerical method. There is a significant difference between the simulation results obtained for the same reference segmentations, see Table 2. The differences can in part be explained by the choice of subsample volumes, as some properties have been obtained as arithmetic averages of up to eight digital subsamples, while other simulated values are obtained using the full-scale model size which was available from the 3-D digital image. This effect is likely to play an important role in the case of upscaling of carbonate results, as this rock sample is particularly sensitive to the choice of subsample volumes.

A second factor that may lead to variations in the simulation results is the choice of boundary conditions. The simulations of electrical properties of the EJ Diffusion method, the FEM-F, the FEM-C and the FDM methods all apply periodic boundary conditions for the electrical current field. Due to the lack of periodicity of the microstructure, boundary artifacts can increase the resistivity of the digital sample since the conducting pore space at the upstream boundary does not match a corresponding porosity at the downstream boundary. In order to reduce this boundary effect, for the FEM-C method, a conducting layer was placed at the downstream boundary, connecting the pores of the downstream to the ones upstream. For the FEM-F method, we have performed a mirroring of the sample geometry in the direction of the mean applied voltage. For the simulation using the FDM method, no special consideration was given to the up- and down-stream boundaries. We speculate that the different choices of boundary conditions in part explain the higher resistivity values determined using the FDM method as compared to the FEM method.

4.3. Subsample size and comparison with laboratory measurements

Ideally, the digital sample size should be large enough to ensure that the properties obtained from numerical simulation are representative for the core from which the digital sample was extracted. This represents a challenge for digital rock physics as the numerical cost of the property prediction grows with the size of the digital image. In order to reduce the computational cost it is therefore common to choose subsamples from a digital image and compute physical properties on those subsamples.

It should be noted that the laboratory measurements presented here are not obtained on the samples that were used to obtain the digital images for our digital rock simulations. Hence, the purpose of the laboratory data is to demonstrate qualitative agreement and similarity in trends. In a qualitative sense, the numerical estimates are in the same range as the laboratory measurements.

We would like to point out that the carbonate rock sample shows a large degree of heterogeneity on the scale of the digital image. Therefore, the numerically estimated rock properties show large variability if the rock is divided into smaller subsamples. As a consequence, the numerical results should be interpreted with care, and multiple digital samples should be analyzed in order to estimate the numerical uncertainty associated with the limited field of view of the digital image.

Computationally obtained moduli for Fontainebleau and Berea sandstones, as well as the sphere pack are consistently stiffer if compared to laboratory measurements or the reference solution from granular dynamics simulation, respectively. This observation is in agreement with previous studies by Arns et al. (2002) or Madadi et al. (2009). A common explanation of this bias of elastic properties is the importance of thin compliant porosity (e.g., Størvoll and Bjørlykke, 2004). Small microcracks are often insufficiently imaged using micro-CT methods and thus underrepresented during image segmentation, specifically in the vicinity of grain contacts. This effect is more pronounced for low-resolution images such as in the case of Fontainebleau sandstone and for soft porous media, such as in the case of the sphere pack.

5. Summary

We numerically predict the effective elastic, hydraulic and electrical properties of four digital rock samples. Three samples are segmented x-ray CT images of Fontainebleau sandstone, of Berea sandstone, and of a vuggy carbonate from the Grosmont formation. The fourth sample is a computer-generated image of a random sphere pack. For the simulation-based prediction of the effective properties, several modeling choices have to be made, such as the choice of image segmentation, subsample size, numerical resolution, the type of the numerical solver, and, finally, the boundary conditions.

The multitude of modeling choices makes this benchmark comparison complex. To the best of our knowledge, this is the first study that presents and analyzes the differences of computed effective properties which result from the various steps in the digital rock physics workflow. Also, this is the first publication that uses the Fourier-transform based Lippmann-Schwinger solver for geomaterials. Further methods that we use to predict effective properties are: the finite-element method, the finite-difference method, the dynamic pulse propagation method for elastic properties, and the explicit jump method.

The predictions of effective properties are affected by several modeling choices: the segmentation process, the choice of the digital subvolume that is used for the simulation, as well as the solver type and boundary conditions. Simulations performed on identical segmented images reveal the importance of the choice of numerical method and the boundary conditions. We provide the range and midrange values of effective properties obtained using the different numerical methods.

With the exception of one simulation, the predicted elastic moduli of natural samples deviate at most 11% from the respective midrange values. We observe up to 20% relative deviation from the midrange value for the bulk modulus of the sphere pack.

As for the hydraulic permeability, the numerically determined values of natural rock samples deviate at most 38% from the midrange value. The best agreement between the numerically predicted values is found for the case of the sphere pack and the Berea sandstone. In both cases, the highest relative deviation is less than 10% of the average permeability.

Finally, the electrical formation factor shows variability comparable to that observed for the hydraulic permeability. In our experiments, the values deviate at most 23% from the respective midrange formation factors. We observe the best agreement for the sphere pack where we predict a range of formation factors between 4.3 and 4.8.

We understand that digital rock studies and laboratory experiments are carried out at different scales. While experimental cores typically have a diameter of several centimeters, the diameter of digital samples is of the order of 2 mm or smaller. Therefore, we should not attempt to directly compare laboratory results with

predictions from digital rock samples. Instead, our results confirm that rock-physics trends such as porosity-permeability relationships are in very good agreement with the corresponding laboratory observations on larger scales.

At this point it is virtually impossible to categorically recommend the preferred solver and setup. Rather, the reader should treat the results as benchmarks and verify newly produced values accordingly.

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